

## Lesson 1.20: Budget Meal Lab

**Unit:** Unit 1, Kitchen Skills and Smart Eating (Nutrition and Wellness) **Topic:** 1.4 Plan, Shop, Cook: Meals on a Budget **Grade:** 6 to 8 (written at grade 7; see Differentiation for the grade 6 support and grade 8 stretch) **Days:** 2 class periods of 40 minutes (Day 1 prep and approval; Day 2 cook, plate, eat, clean) **Room:** FACS lab with three kitchen stations (a no-kitchen version is below)

### Standards

- NYS module line: NYS N&W 1. Culinary Management b) Demonstrate safety and sanitation procedures when handling food and equipment
- NYS module line: NYS N&W 1. Culinary Management d) Prepare a simple recipe
- NYS module line: NYS N&W 1. Culinary Management g) Prepare and serve a simple meal
- NYS module line: NYS N&W 1. Culinary Management h) Demonstrate appropriate table setting
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment
- CTE theme line: CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace
- CTE theme line: CTE CIR 3. Workplace Communication f) Demonstrate effective communication skills in a group setting to accomplish a task
- NYS FACS Learning Standard 1 (Intermediate), Key Idea: Students will use an understanding of the elements of good nutrition to plan appropriate diets for themselves and others. They will know and use the appropriate tools and technologies for safe and healthy food preparation.; Performance Indicator: Students understand the relationships among diet, health, and physical activities; evaluate their own eating patterns; and use appropriate technology and resources to make food selections and prepare simple, nutritious meals.; Performance Indicator: Students apply principles of food safety and sanitation.
- CDOS Standard 3a Universal Foundation Skills: Interpersonal Skills
- National FCS 3.0: 14.3.3 Show the ability to select, store, prepare, and serve food that is nourishing and looks good on the plate.

(Four module lines are cited because this is the unit's summative lab and all four are scored on Rubric 01, criterion 3.)

### Enduring understanding and essential question

**Big idea:** A plan on paper becomes a meal only when seven people follow one card, do their own job, and keep the station safe and clean from the first minute to the last. **Essential question:** What has to be true at the station before a team can cook, and what has to be true before it can leave?

## Objectives

By the end of this lesson, students will be able to:

1. (Low) Given the team's plan, write a station card (ingredients scaled for seven tasting portions, eight to ten steps, roles, a five-minute-block flowchart, safety reminders, and a "what does done look like" line) that the teacher can approve.
2. (Mid) Working in a role at a station, prepare, plate, and serve the team's dish in 40 minutes following the safety and sanitation rules of the Kitchen Safety Contract, with the station clean and checked at the bell.
3. (High) At the end of Day 2, evaluate the team's own lab on the lab rubric (safety, the dish, teamwork) and write one change the team would make if it cooked the dish again.

## Vocabulary

Tier 2 (general academic): scale, sequence, approve, evaluate Tier 3 (FACS): station card, flowchart, tasting portion, mise en place (everything in its place before cooking), place setting, sanitize, leftovers. See the unit vocabulary list for the Spanish column.

## Materials and setup

- Handout, one per team: Handout 01.20 - Team Station Card Template.md
- Handout, three copies for the teacher and one per team that needs one: Handout 01.20 - Fallback Station Cards.md (three costed dishes that meet every constraint)
- Each team's revised and signed Project Plan Sheet from Lesson 1.19
- Lab role cards, one set of seven per station: Station Manager, Prep Lead, Cook Lead, Plate Captain, Clean Crew (two), Safety and Sanitation Checker
- Groceries for each team's approved dish, bought after Day 1, budgeted at \$20 per station and three stations per period
- Station equipment: one pot, one large skillet or sheet pan, colander (teacher use), mixing bowl, two forks for shredding, measuring cups and spoons, wooden spoon, spatula, oven mitts, cutting board (for assembly only, no knives), sanitizer spray and paper towels, dish soap, food thermometer, foil, paper plates or reusable plates, forks, spoons, napkins, cups
- Timer visible at each station
- Lab rubric on the slide and one printed copy per team for the self-assessment
- Slides: Slides 01.20 - Budget Meal Lab (outline in 05 Materials; the site renders it as a deck)
- Setup notes: Day 1 needs no food. After Day 1, shop from the approved station cards or the fallback cards. On Day 2, preheat the oven before the first period and stage each station's groceries in a labeled bin with the station card on top. Allergy check: on Day 1, read every approved card against the class allergy list; any team whose dish touches a listed allergen takes a fallback card or swaps the ingredient. The chicken quesadilla fallback contains dairy and wheat; the pasta fallback contains wheat and dairy; the rice and bean bowl contains dairy (the cheese and sour cream can be left off one portion).

## Pre-assessment

Every student passed the Kitchen Safety Exam (Lesson 1.4) and has cooked twice (Lessons 1.8 and 1.13). The Day 1 do now asks what went wrong last time; the answers tell you which safety reminders go on every card.

## Do now and hook (Time: 5 min, Day 1)

On the board: “Think about Lab 2 (the nachos). Write one thing your station did well and one thing that went wrong or ran late.”

Debrief: collect the “went wrong” answers on the board. Most will be about time and cleanup. Then the hook: “This time the recipe is yours. Nobody handed you a card. You write it, I sign it, I shop it, and on Day 2 you cook it. If the card is wrong, the dish is wrong. Today is about getting the card right.”

## Procedure

| Step                                                          | Teacher will                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Students will                                                   | Time   |
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| Day 1, 1. Direct instruction: what a station card has to have | Show a filled fallback card on the screen (the pasta one). Point at the seven parts: ingredients scaled for seven tasting portions with the package size and cost, eight to ten numbered steps, seven roles with a job per phase, a flowchart in five-minute blocks (prep 0 to 5, cook 5 to 25, plate and eat 25 to 32, clean 32 to 40), the constraints checklist (no knives, pre-cooked meat only, stovetop and oven, teacher drains and handles the oven), safety reminders, and “what does done look like.” | Read the fallback card. Answer: “Which step needs the teacher?” | 6 min  |
| Day 1, 2. Learning activity: write the                        | Teams write their card from the station                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Write the card: ingredients and cost,                           | 18 min |

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| team station card                    | dish on their plan. Scaling rule: the family plan feeds four full servings; the station makes seven tasting portions, which is about the same amount of food, so most quantities stay the same. The Materials Manager and Shopper total the station ingredients against \$20 with a unit on every line. Circulate; check the constraints checklist first, the cost second, the steps third. | steps, roles, flowchart, safety, done line.                                                              |       |
| Day 1, 3. Approval and shopping list | Sign each card that passes: constraints met, \$20 or under, allergen check clear, steps doable in 20 minutes of cooking. A card that fails gets one specific fix and five minutes; if it still fails, the team takes the fallback card closest to its dish and marks on the plan why. Collect the approved cards; they are the shopping list.                                               | Fix and resubmit, or choose a fallback. Assign the seven lab roles for Day 2 and write them on the card. | 7 min |
| Day 1, closure                       | See Closure.                                                                                                                                                                                                                                                                                                                                                                                | See Closure.                                                                                             | 4 min |
| Day 2, do now (clock 0 to 2)         | Students wash hands, put on aprons, and tie back hair as they enter (the routine from Labs 1 and 2). On the board: "Read your station card. Write the one step you are                                                                                                                                                                                                                      | Wash, apron, hair back, read, write one line on the back of the card.                                    | 2 min |

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|                                         | most likely to get wrong and what you will do about it.”                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                            |        |
| Day 2, 4. Lab: prep (clock 2 to 5)      | Give the reset cue (lights, countdown, hands empty) and the three reminders: the oven and the colander are mine; no knives exist today; the Safety Checker calls the two-hour rule at cleanup. Start the timers. | Station Manager reads the card aloud. Prep Lead sets out every ingredient and tool (mise en place). Safety Checker checks the fridge temperature and that every hand was washed. Plate Captain sets one place setting per person: plate, fork on the left, spoon on the right, napkin under the fork, cup above the spoon. | 3 min  |
| Day 2, 5. Lab: cook (5 to 25)           | Circulate between the three stations with the lab rubric on a clipboard. Drain pasta and open the oven when called. Ask the “what does done look like” question at each station once.                            | Cook Lead runs the stovetop or assembles for the oven. Prep Lead hands ingredients in order. Station Manager watches the flowchart and calls time. Clean Crew washes as they go. Safety Checker takes the temperature of any reheated meat (165 F) and watches for towels on burners and handles turned out.               | 20 min |
| Day 2, 6. Lab: plate and eat (25 to 32) | Say “plate.” Check each plate for the family it was planned for: does it look like the menu card will say it does? Sit and eat with one station.                                                                 | Plate Captain plates seven tasting portions. Everyone sits and eats at the set table. Presenter notes how it looks and tastes for the menu card                                                                                                                                                                            | 7 min  |

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|                                                   |                                                                                                                                                                                                                                                                                                                                                       | tomorrow.                                                                                                                                                                                            |       |
| Day 2, 7. Lab: clean and station check (32 to 40) | Run the station check with the Station Manager at each station: surfaces sanitized, tools washed and put away, burners off and cool, sink empty, floor dry, leftovers in a labeled container in the fridge or in the trash (the Safety Checker says which and why). Sign the card. Teams fill the lab rubric self-assessment on the back of the card. | Clean Crew leads; everyone cleans. Safety Checker handles leftovers by the two-hour rule and labels the container with the date. Team fills the self-assessment and writes one change for next time. | 8 min |

Day 1 totals 40 minutes: 5 + 6 + 18 + 7 + 4. Day 2 totals 40 minutes: 2 + 3 + 20 + 7 + 8 (the do now and prep share the 0 to 5 block on the station clock; the lab rubric self-assessment in Step 7 is the closure).

### Questions to ask

Monitor understanding:

- What is the first thing on the flowchart, and who does it?
- What temperature does reheated chicken have to reach? How do you know it got there?
- Your leftovers are on the counter at 32 minutes. Where are they at 40, and why?

Deepen learning:

- The card says “stir until the sauce bubbles.” What does bubbling tell you that a timer cannot?
- Your dish cost \$17 for seven tasting portions. What does that make the cost of one portion? Is that more or less than the \$40 plan’s cost per person?
- Which role was the hardest today? Would you take it again? Why or why not?

### Check for understanding

Day 1: clipboard yes or no per team on “constraints checklist all checked,” “station total at or under \$20 with a unit on every line,” and “flowchart has a name in every block.” Day 2: the lab rubric on the clipboard during Step 5, one line per station: safety and sanitation (hands, surfaces, handles, temperature), the dish (followed the card, plated as planned), teamwork (every role working, station manager calling time). A “got it” station has no safety reminder from the teacher during the cook phase.

## Closure (Time: 4 min Day 1; folded into Step 7 Day 2)

Day 1: exit card, 3 parts of your station card, 2 things the teacher checked before signing, 1 step you are nervous about. Two students read the nervous step aloud; those become the Day 2 reminders. Day 2: the self-assessment on the back of the station card (three objectives plus the safety and sanitation row, 1 to 4 each) and one sentence: "If we cooked this again, we would \_\_\_\_." The Station Manager reads the sentence aloud as the team leaves.

## Differentiation and supports

- ELL: the station card template has an icon per section and a filled example (the fallback card) to copy from; the seven role cards have a picture and a one-line job; the flowchart is visual; the place setting is a diagram on the slide; sentence starter for the change line: "Next time we would \_\_\_\_ because \_\_\_\_"; the words stir, boil, drain, plate, sanitize, and leftovers are on the unit vocabulary card in Spanish.
- IEP and 504 (general): choice of role (Prep Lead, Plate Captain, and Clean Crew have clear physical tasks and no time pressure at the stove); the card template has the flowchart blocks pre-drawn; a student who cannot stand at a burner is the Safety and Sanitation Checker with the thermometer and the clipboard; extended time on the card by finishing during the Lesson 1.19 revision window; a written lab alternative (a recipe analysis of the team's card) for any student absent on Day 2, per the grading plan.
- Grade 6 support: the team takes a fallback card and edits it (changes two ingredients or two steps to match its plan) instead of writing a card from blank; the flowchart is filled in as a class.
- Grade 8 stretch: the card includes a cost per tasting portion and a cost per full serving, a sales tax line for the non-food items, and a critical path (the one chain of steps that cannot slip without the dish being late), which is the Gantt idea from my lesson packages.
- UDL checkpoint used: multiple means of action and expression (seven roles, each a different way to do the lab well) and multiple means of representation (the card as text, as a flowchart, and as a filled example).
- No-kitchen or no-machine alternative: the team builds its plated dinner from the food cards (the Unit 1 food card set, one card per ingredient with a picture and a price) on a paper plate template, arranged by MyPlate section, with the costed plan beside it. Then the team assembles a no-cook component for real: a salad (bagged greens, canned beans, shredded cheese, bottled dressing, no knives) or a cold wrap (tortilla, pre-cooked deli chicken or beans, cheese, salsa) using the same roles, the same flowchart, the same place setting, and the same clean and station check. Cost per station for the no-cook version is about \$12. Rubric 01 criterion 3 is scored on the no-cook assembly.

## Assessment

| Objective                           | Standards met                   | Learning task                     | Assessment (formative or summative; tool)                                        |
|-------------------------------------|---------------------------------|-----------------------------------|----------------------------------------------------------------------------------|
| 1. Write an approvable station card | N&W 1. d); HSW 4. c); CIR 3. f) | Station card, Day 1 Steps 2 and 3 | Formative; teacher sign off on the card (constraints, cost, steps); feeds Rubric |

|                                             |                                                                                                           |                               |                                                                                                                                                                                           |
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|                                             |                                                                                                           |                               | 01 criterion 1                                                                                                                                                                            |
| 2. Prepare, plate, serve, and clean in role | N&W 1. b), d), g), h); HSW 4. c), e); Std 1 PI A and B; National FCS 14.3.3; CDOS 3a Interpersonal Skills | Lab, Day 2 Steps 4 to 7       | Summative; lab rubric (three objectives plus a safety and sanitation row, 1 to 4 each, 16 points, per the grading plan) and Rubric 01 criterion 3 (the cooked dish, safety, and teamwork) |
| 3. Self-assess and propose one change       | CIR 3. f); CDOS 3a Interpersonal Skills                                                                   | Self-assessment, Day 2 Step 7 | Formative; self-assessment compared to the teacher's lab rubric; the change sentence goes on the menu card if the team wants it                                                           |

## Homework

None. Optional: cook the team's dish at home for your family and report the real cost.

## Connections

Math: cost per portion and scaling are ratio work. Science: heat, boiling, and the 165 F rule are the science teacher's thermal energy and microbe lessons in one pot. Community: a local restaurant cook or the school cafeteria manager watching the cook phase and giving one piece of feedback per station is a fifteen-minute visit that students remember.

## Sources

- NYS Middle Level CTE FACS Content Module 05, Nutrition and Wellness (nycotecenter.org, June 2018), lines 1. b), 1. d), 1. g), and 1. h). Copy in 02 Standards/NYS Module PDFs/.
- NYS Middle Level CTE Theme Module 04, Health, Safety, and Wellness, lines 4. c) and 4. e); Theme Module 02, Communication and Interpersonal Relationships, line 3. f). Same folder.
- NYS Learning Standards for Health, Physical Education, and Family and Consumer Sciences (1996), Standard 1, Intermediate. Wording per 02 Standards/STANDARDS CROSSWALK.md, Part A.6.
- CDOS Intermediate Standard 3a, per 02 Standards/STANDARDS CROSSWALK.md, Part A.7.
- National Standards for Family and Consumer Sciences Education 3.0 (LEAD FCS Education, 2018), competency 14.3.3, per 02 Standards/STANDARDS SOURCES - web research.md.

- USDA safe minimum internal temperature for reheated leftovers and poultry (165 F), FoodSafety.gov.
- My own materials: the supermarket station cards, master flowchart, and menu options: the seven-student station, the constraints line, the “what does done look like” checklist, the teacher-drains-pasta rule, and the rotisserie chicken shredded with two forks. The role card structure (core roles plus expansion roles) and the PREP, COOK, PLATE and EAT, CLEAN timeline from my three FACS Cooking Lesson Packages. The Budget Meal Lab idea from my draft unit calendar.

### Teacher notes

- If Day 1 runs short, the flowchart can be the do now on Day 2 (five minutes, the Station Manager fills it while the others wash hands). If Day 2 runs short, plate at 27 and eat standing; never cut the clean and station check.
- The fallback cards exist so no team goes hungry because its plan failed. A team on a fallback still gets full credit on criterion 3; it loses on criterion 1 only if its plan was what failed.
- Common mistake: teams write “cook the chicken” on the card. The chicken is pre-cooked. The step is “heat the chicken until the thermometer reads 165 F.” Fix it at approval.
- Second common mistake: the Cook Lead does everything and five people watch. The flowchart with a name in every block is the fix, and the lab rubric’s teamwork line scores it.
- Leftovers: the Safety Checker decides by the rules from Lesson 1.17. Anything held above 40 F for over two hours by the end of the day is trash, not tomorrow’s tasting gallery. If the school day makes same-day refrigeration impossible, plan a fresh no-cook item for Lesson 1.21 instead (see that lesson).
- Cost estimate per station for the three fallback dishes (prices for a Long Island supermarket, September 2026, every one on the cards, to be refreshed from the current circular): rice and bean bowl about \$13.33; pasta with vegetables and jarred sauce about \$11.05; pre-cooked chicken quesadillas about \$17.64. All three under the \$20 cap. One period with one of each is about \$42.02; per-period cost with three teams on their own dishes will vary but is capped at \$60.